

Theory of Automata

Midterm Solution

Question 1

Part (a)

[Total: 3 Marks]

Problem: Consider the language S^* , where $S = \{ab, ba\}$. Write out all the words in S^* that have seven or fewer letters. Can any word in this language contain the substrings aaa or bbb ?

Explanation & Solution: The language S^* (Kleene Star of S) is formed by concatenating the "building blocks" from set S (ab and ba) as many times as needed.

- Since both blocks (ab and ba) have a length of exactly 2, any word formed by joining them must have an **even length** ($0, 2, 4, 6 \dots$).
- Therefore, it is impossible to have words of odd lengths like 1, 3, 5, or 7.

[1 Mark]

1. List of words (Length ≤ 7):

- **Length 0:** λ (The empty string, which is always in S^*).
- **Length 2:** The single blocks: ab, ba .
- **Length 4:** Combinations of 2 blocks ($2 \times 2 = 4$ words):

$abab, abba, baab, baba$

- **Length 6:** Combinations of 3 blocks ($4 \text{ words} \times 2 \text{ blocks} = 8 \text{ words}$):

$ababab, ababba, abbaab, abbaba$

$baabab, baabba, babaab, bababa$

[1 Mark]

2. Can it contain aaa or bbb ? To form aaa , we would need three consecutive identical letters. Let's look at the boundaries where blocks join:

- Every a in the blocks is adjacent to a b .

- The maximum number of identical consecutive characters occurs when we join $ab + ba$ (producing $a\mathbf{b}b\mathbf{a}$) or $ba + ab$ (producing $b\mathbf{a}a\mathbf{b}$).
- In both cases, the run of identical letters is only 2.

Answer: No.

[1 Mark]

Part (b)

[Total: 4 Marks]

Problem: Prove that there are as many palindromes of length $2n$ as there are of length $2n - 1$ ($n = 1, 2, 3$).

Explanation & Proof: A palindrome is symmetric. The number of possible palindromes depends on how many "free choices" we have before the rest of the string is forced to be a mirror image.

1. **Even Length ($2n$):** The string looks like $x_1x_2 \dots x_nx_n \dots x_2x_1$. We choose the first n characters. Once these are picked, the second half is strictly determined.

$$\text{Total Choices} = |\Sigma|^n = 3^n$$

[1.5 Marks]

2. **Odd Length ($2n - 1$):** The string looks like $x_1 \dots x_{n-1}\mathbf{M}x_{n-1} \dots x_1$. We choose the first $n - 1$ characters (the "left wing") AND the 1 middle character (**M**).

$$\text{Total Choices} = |\Sigma|^{n-1} \times |\Sigma|^1 = |\Sigma|^n = 3^n$$

[1.5 Marks]

Since both calculations result in 3^n , the number of palindromes is identical.

n	Even Len ($2n$)	Count (3^n)	Odd Len ($2n - 1$)	Count (3^n)
1	Length 2 (aa, bb, cc)	$3^1 = 3$	Length 1 (a, b, c)	$3^1 = 3$
2	Length 4	$3^2 = 9$	Length 3	$3^2 = 9$
3	Length 6	$3^3 = 27$	Length 5	$3^3 = 27$

[1 Mark]

Answer: The number of palindromes of length $2n$ is 3^n .

Question 2

Problem: Determine the Regular Expression (RE) over $\Sigma = \{a, b\}$.

a. Beginning with a [Total: 3 Marks]

Explanation: The first letter is fixed as a . The remainder of the word can be any sequence of a 's and b 's (including empty), which is represented by $(a + b)^*$. [1 Mark]

$$\mathbf{RE} : a(a + b)^*$$

[2 Marks]

b. Not ending in a [Total: 3 Marks]

Explanation: If a word does not end in a , it implies it must end in b . We must also include the empty string λ because it technically does not end in a either. [1 Mark]

$$\mathbf{RE} : (a + b)^*b + \lambda$$

[2 Marks]

c. Beginning and ending in the same letter [Total: 3 Marks]

Explanation: We must cover three cases:

1. Words starting and ending with a (minimum length 2): $a(a + b)^*a$
2. Words starting and ending with b (minimum length 2): $b(a + b)^*b$
3. Single letters (start and end with themselves): $a + b$

[1 Mark]

$$\mathbf{RE} : a(a + b)^*a + b(a + b)^*b + a + b$$

[2 Marks]

Question 3

Problem: Give a Recursive definition for:

a. The set $ODD = \{1, 3, 5, 7 \dots\}$ [Total: 3 Marks]

Explanation: We start with the first odd number (1). Any subsequent odd number is found by adding 2 to the previous one.

• **Base Case:** $1 \in ODD$ [1.5 Marks]

• **Recursive Step:** If $x \in ODD$, then $x + 2 \in ODD$. [1.5 Marks]

b. Strings of digits $0 \dots 9$ that cannot start with 0 [Total: 3 Marks]

Explanation: We first establish the valid starting digits (1 through 9). Once we have a valid number, appending any digit (0-9) to the end results in a new valid number (e.g., if 5 is valid, 50 is also valid). Let S be the valid set.

- **Base Case:** $1, 2, 3, 4, 5, 6, 7, 8, 9 \in S$ [1.5 Marks]
- **Recursive Step:** If $w \in S$, then $w0, w1, \dots, w9 \in S$. [1.5 Marks]

c. The language AA of words containing substring aa [Total: 3 Marks]

Explanation: The simplest word satisfying the condition is "aa". If a word already contains "aa", adding any letters to the front or back does not remove the "aa"; the word remains in the language.

- **Base Case:** $aa \in AA$ [1.5 Marks]
- **Recursive Step:** If $w \in AA$, then $aw, bw, wa, wb \in AA$. [1.5 Marks]